ANALYSIS REPORT

Track 1 Remaining Family Final Closeout Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the full remaining-family exact-paper `Track 1` batch prepared in:

- `doc/reports/campaign_plans/track1/exact_paper/2026-04-18-00-48-05_track1_remaining_exact_paper_family_campaigns_plan_report.md`

  The earlier partial closeout already covered the first seven completed families. This final report promotes the recovered `XGBM` and `LGBM` reruns into the canonical campaign bookkeeping and removes the temporary `pending rerun` state from the batch.
- completed family campaigns: `9`
- completed validation runs: `18`
- failed validation runs: `0`
- aggregate campaign artifact root:

  `output/training_campaigns/track1_remaining_family_full_matrix_campaigns_2026_04_18_00_48_05/`

Operational Outcome

- the recovered `XGBM` rerun completed both amplitude and phase exact-paper runs
- the recovered `LGBM` rerun completed both amplitude and phase exact-paper runs
- the `XGBM` and `LGBM` family campaign folders now expose:
  - `campaign_leaderboard.yaml`
  - `campaign_best_run.yaml`
  - `campaign_best_run.md`
- the aggregate remaining-family campaign root now exposes the final `18`-entry

  leaderboard and the final winning run
- `doc/running/active_training_campaign.yaml` can now move from

  `interrupted` to `finished`

Family Recovery Outcome

Family	Amplitude Run	Phase Run	Best Run	Best Scope	Best Closure Score	Open Cells
XGBM	track1_xgbm_amplitude_full_matrix	track1_xgbm_phase_full_matrix	track1_xgbm_phase_full_matrix	phases_only	0.111	14
LGBM	track1_lgbm_amplitude_full_matrix	track1_lgbm_phase_full_matrix	track1_lgbm_amplitude_full_matrix	amplitudes_only	0.550	4

Aggregate Ranking

Rank	Run	Family	Scope	Paper Cell	Met	Near	Open	Closure Score
1	track1_rf_phase_full_matrix	RF	phases_only	18	7	8	3	0.611
2	track1_gbm_phase_full_matrix	GBM	phases_only	18	7	6	5	0.556
3	track1_hgbm_phase_full_matrix	HGBM	phases_only	18	6	8	4	0.556
4	track1_lgbm_amplitude_full_matrix	LGBM	amplitudes_only	10	5	1	4	0.550
5	track1_lgbm_phase_full_matrix	LGBM	phases_only	18	6	7	5	0.528
6	track1_ert_phase_full_matrix	ERT	phases_only	18	7	3	8	0.472
7	track1_hgbm_amplitude_full_matrix	HGBM	amplitudes_only	10	3	3	4	0.450
8	track1_ert_amplitude_full_matrix	ERT	amplitudes_only	10	2	3	5	0.350
9	track1_rf_amplitude_full_matrix	RF	amplitudes_only	10	1	5	4	0.350
10	track1_dt_phase_full_matrix	DT	phases_only	18	2	8	8	0.333
11	track1_gbm_amplitude_full_matrix	GBM	amplitudes_only	10	1	4	5	0.300
12	track1_et_phase_full_matrix	ET	phases_only	18	2	1	15	0.139
13	track1_xgbm_phase_full_matrix	XGBM	phases_only	18	0	4	14	0.111
14	track1_dt_amplitude_full_matrix	DT	amplitudes_only	10	0	2	8	0.100
15	track1_et_amplitude_full_matrix	ET	amplitudes_only	10	0	2	8	0.100
16	track1_xgbm_amplitude_full_matrix	XGBM	amplitudes_only	10	0	1	9	0.050
17	track1_mlp_amplitude_full_matrix	MLP	amplitudes_only	10	0	0	10	0.000
18	track1_mlp_phase_full_matrix	MLP	phases_only	18	0	0	18	0.000

Final Batch Winner

The aggregate batch winner remains `track1_rf_phase_full_matrix` from family `RF`.

The recovered `XGBM` and `LGBM` reruns materially complete the batch but do not displace the existing aggregate winner under the explicit closure-first ranking policy.

- winning scope: `phases_only`
- closure score: `0.611`
- met paper cells: `7`
- near paper cells: `8`
- open paper cells: `3`

Scientific Outcome

- `XGBM` is now fully validated as a completed row, but it does not improve the canonical benchmark envelope materially
- `LGBM` closes the batch with a strong amplitude row and a competitive phase row
- the main final benchmark gains versus the earlier partial closeout come from `LGBM`:
  - harmonic `1` amplitude `MAE` moves from near-target to met-target
  - harmonic `3` amplitude `RMSE` moves from near-target to met-target
- the aggregate winner and the highest-priority open harmonics remain unchanged

Final Benchmark Refresh Effect

The canonical benchmark after the full remaining-family closeout is:

- Table `2` amplitude `MAE`: `3/10` harmonics meet or beat the paper target
- Table `3` amplitude `RMSE`: `6/10` harmonics meet or beat the paper target
- Table `4` phase `MAE`: `5/9` harmonics meet or beat the paper target
- Table `5` phase `RMSE`: `5/9` harmonics meet or beat the paper target
- harmonic-level Table `6` closure: `2/10` fully matched, `5/10` partially matched, `3/10` not yet matched

The highest-priority still-open harmonics remain:

- `3`
- `81`
- `156`
- `162`
- `240`

Final Interpretation

This batch is now operationally complete and scientifically much better anchored than the interrupted partial-closeout state.

The key final conclusion is

- the remaining-family batch is fully closed out;
- `RF phase` remains the strongest bookkeeping winner;
- `LGBM` improves the amplitude-side paper envelope;
- `Track 1` still remains canonically open because the exact-paper tables are not yet fully matched at harmonics `3`, `81`, `156`, `162`, and `240`.